

Worked for 32s

Yes. I've inspected the ZIP and it contains a substantial codebase (2,159 files) including the PHP engine, knowledgebase, documentation, validation framework, reports, prototypes and test personas.

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The right deliverable is not a chat summary. It should become the **Project Architecture Document** that lives with the repository and serves as the onboarding reference for any future ChatGPT session or developer.

Personal Wellness Blueprint (PWB)

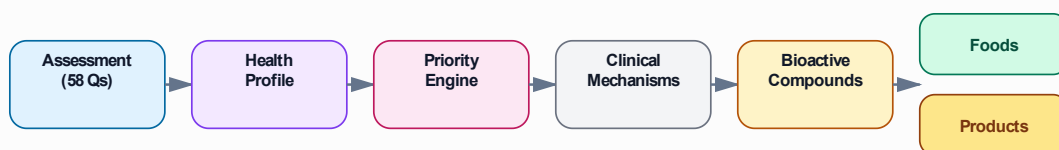
Project Architecture & Technical Overview v1.0

Status: Explainable Clinical Nutrition Recommendation Engine

1. Executive Summary

The **Personal Wellness Blueprint (PWB)** is a PHP-based clinical nutrition recommendation platform that converts a structured health questionnaire into personalised nutritional recommendations through an entirely explainable decision pipeline.

Unlike AI systems that generate recommendations from opaque models, every recommendation in PWB is traceable through explicit clinical relationships:



Every stage is independently testable and produces auditable outputs rather than hard coded recommendations. This explainable architecture is already documented within the repository and forms the core design principle of the engine.

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2. Project Objectives

Primary objective

Create a scientifically grounded recommendation engine that produces:

- Personalised food recommendations
- Bioactive compound priorities
- Nutritional supplement recommendations
- Fully explainable clinical reasoning
- Professional PDF wellness reports

Design principles

- Explainability over black-box AI
- Evidence-based taxonomy
- Deterministic scoring
- Modular PHP architecture
- JSON-driven knowledgebase
- Clinician-readable outputs

3. Repository Structure

The ZIP contains the following high-level architecture:



knowledgebase/

Canonical clinical data

engine/

PHP recommendation engine

docs/

Technical documentation

assets/

UI resources

output/

Generated reports

tools/

Utilities & generators

prototypes/

Experimental interfaces

The architecture is already well separated between **data**, **engine**, **reports**, and **validation**, making future maintenance considerably easier.

4. Knowledgebase Architecture

The knowledgebase is the heart of the project.

It contains structured JSON taxonomies rather than embedded PHP logic.

Current domains

- | | |
|--|---|
| ✓ Foods
Individual JSON monographs | ✓ Vitamins
Nutrient database |
| ✓ Minerals
Nutrient database | ✓ Body systems
Clinical domains |
| ✓ Assessment questions
58-question scoring model | ✓ Products
Supplement catalogue |
| ✓ Personas
Test assessments | ✓ Graph relationships
Mechanism mapping |

The repository already stores each food as its own JSON document (`FD001-oats.json` , `FD002-blueberries.json` , etc.), indicating the intended scalable architecture rather than one enormous data file.

5. Explainable Recommendation Engine

The repository implements the recommendation pipeline as discrete resolver classes.

Engine classes

Class	Responsibility
<code>HealthProfileBuilder</code>	Converts questionnaire into body-system scores
<code>PriorityEngine</code>	Ranks clinical priorities
<code>MechanismResolver</code>	Converts priorities into mechanisms
<code>BioactiveResolver</code>	Scores therapeutic compounds
<code>FoodResolver</code>	Ranks foods
<code>ProductResolver</code>	Matches supplements
<code>RecommendationEngine</code>	Orchestrates the pipeline

These components are explicitly listed within the project's own engine state documentation.

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Orchestration

The `RecommendationEngine` demonstrates the intended deterministic workflow:

1. Calculate priorities
2. Score bioactives
3. Score foods
4. Return explainable results

The implementation contains no embedded clinical rules; it delegates each stage to dedicated resolver classes.

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6. Assessment System

The questionnaire contains **58 structured questions**.

Rather than assigning scores directly in PHP, the `HealthProfileBuilder` loads the scoring matrix from JSON:

```
question-body-system-scoring.json
```

This means the questionnaire can be modified without altering engine code. The builder normalises single-choice and multi-select answers before accumulating weighted body-system scores.

Current body systems

The engine presently scores nine primary systems:

ID	Domain
BS001	Digestive
BS002	Cardiovascular
BS003	Metabolic

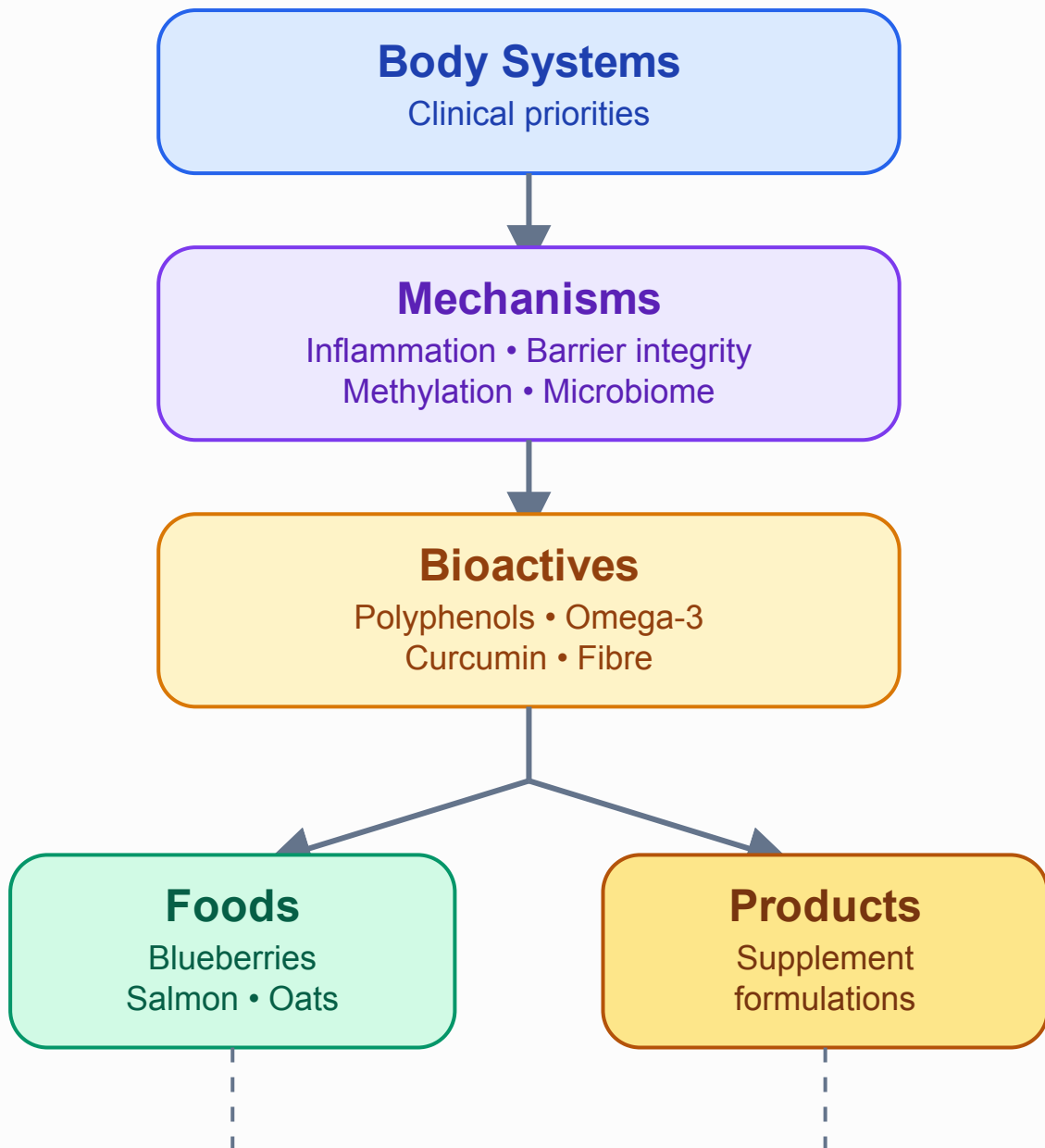
BS004	Immune
BS005	Cognitive
BS006	Musculoskeletal
BS007	Hormonal
BS008	Respiratory
BS009	Detoxification

These IDs are visible within the scoring builder and act as stable identifiers throughout the engine.



7. Clinical Taxonomy Model

The clinical model is intentionally relational.



Every recommendation remains
fully traceable

No direct Body System → Food links

The key architectural decision is that **foods are never linked directly to body systems**.

Instead:

Body System → Mechanism → Bioactive → Food/Product

This creates much greater clinical flexibility because a single mechanism (e.g. reducing oxidative stress) may contribute to multiple body systems.

8. Food Mechanism Audit (Current Major Project)

The largest ongoing task is replacing legacy food mappings with a clinically audited mechanism matrix.

Original situation

Foods contained inconsistent mechanism assignments derived from multiple historical sources.

Problems included:

- duplicated mechanisms
- inconsistent terminology
- overlapping biological concepts
- difficult maintenance
- impossible validation

New architecture

A canonical taxonomy has been designed:

Layer	Purpose
Body Systems	Clinical domains
Mechanisms	Biological processes
Foods	Whole foods
Matrix	Food → Mechanism relationships

This becomes the **single source of truth** from which JSON, Excel and PHP structures will all be generated.

This is the work we have been performing over the recent audit sessions.

9. Validation Framework

One of the strongest aspects of the repository is its dedicated validation subsystem.

Validators

- SchemaValidator

- TaxonomyValidator
- RepositoryValidator
- ValidationEngine

These exist independently of the recommendation engine and allow data integrity to be verified before deployment.

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Intended validation rules

The current project direction should enforce:

- Unique IDs
- Referential integrity
- Valid mechanism references
- No orphan nodes
- Schema compliance
- Duplicate detection

This should run automatically whenever taxonomy JSON is regenerated.

10. Testing Infrastructure

The repository already includes dedicated clinical testing tools.

Assessment

- Assessment loader
- Minimum profile
- Maximum profile
- Persona loader

Clinical

- Priority engine tests
- Mechanism resolver tests

Nutrition

- Food resolver
- Explainability tests

Bioactives

- Bioactive resolver
- Explanation validation

Products

- Product resolver
- Product explanation tests

The documentation explicitly identifies these test suites as part of the explainable engine workflow.

11. Reporting System

The report generator is modular.

Sections

- Body Systems
- Foods
- Bioactives
- Action Plan
- Mechanism Popovers

The inclusion of `MechanismPopover.php` indicates the UI has already begun moving towards contextual clinical explanations rather than static reports.



This aligns with yesterday's design discussion about allowing users to click or hover over mechanisms for deeper explanations.

12. Original Data Sources

The project has evolved from several evidence sources rather than a single database.

Primary sources

Source	Use
Published nutrition literature	Food bioactives

Functional medicine concepts	Mechanism taxonomy
Nutrient composition databases	Vitamins & minerals
Existing product catalogue	Supplement mapping
Custom clinical design	Questionnaire & scoring

Importantly, the repository separates **scientific knowledge** from **clinical relationships**, allowing evidence to be updated without rewriting engine code.

Current Project State

✓ Completed

- ✓ 58-question assessment framework
- ✓ Priority engine
- ✓ Bioactive resolver
- ✓ Product resolver
- ✓ Validation framework
- ✓ Body system scoring engine
- ✓ Mechanism resolver architecture
- ✓ Food resolver
- ✓ Explainable report architecture

🕒 In Progress

- 🕒 Canonical Food → Mechanism matrix
- 🕒 Excel matrix generation
- 🕒 Versioned taxonomy release process
- 🕒 Taxonomy JSON generator
- 🕒 Interactive mechanism explanations

Recommended next repository addition

I recommend adding this as:

📄 [/docs/PWB_PROJECT_ARCHITECTURE_v1.0.md](#)

📖 [/docs/README.md](#)

Project entry point linking to architecture, schema, roadmap and coding standards.

This document is sufficiently comprehensive that a new ChatGPT session could use it as the primary onboarding reference, greatly reducing reliance on conversational history.